Voltage Protector - Simulation Screenshot Records

Wokwi simulation of the STM32F103 (Blue Pill) voltage protector firmware. Defaults: OV 250 V, UV 170 V, OC 10 A. Pot 1 = voltage, pot 2 = current; the relay is the LED on PB0, and the LCD and UART [STATUS] lines show state. Each scenario below matches a file in test_evidence/scenarios; the screenshots follow in the order listed. Scenario 08 (setting takes effect) has no screenshot in this report.

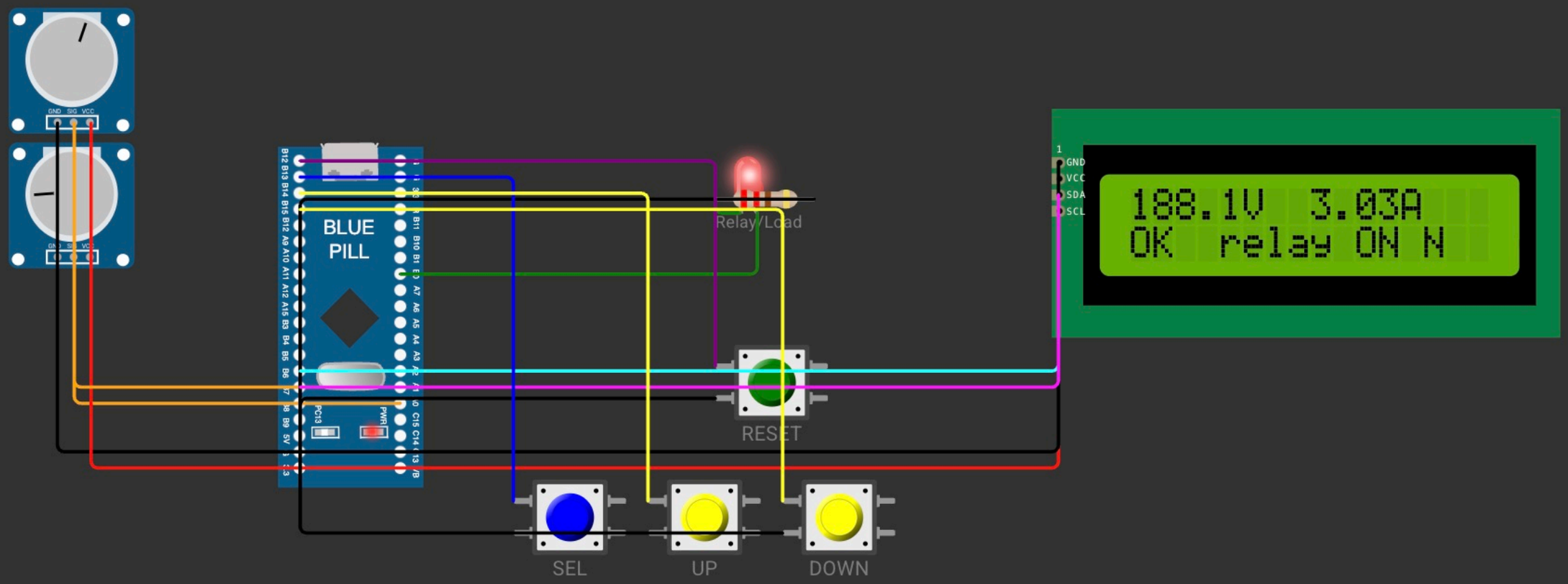
Screenshot	Scenario	Stimulus	Expected	Observed output / result
PDF p.2 01_normal_operation.png	01 Normal operation	Both pots mid-range; voltage inside the 170-250 V window, current below 10 A.	MONITORING / NONE, relay on.	188.1 V, 3.03 A. LCD: OK / relay ON. Terminal: STATE:MONITORING FAULT:NONE, OV:250V UV:170V OC:10.0A. Relay LED lit. PASS.
PDF p.3 02_overvoltage_trip.png	02 Overvoltage	Voltage pot raised above the 250 V limit.	DISCONNECTED / OVERVOLTAGE, relay off.	321.3 V. LCD: TRIP OV / relayOFF. Terminal: STATE:DISCONNECTED FAULT:OVERVOLTAGE. Relay LED off. PASS. (One status line shows a garbled OV field, a serial glitch; the next lines are correct.)
PDF p.4 03_undervoltage_trip.png	03 Undervoltage	Voltage pot lowered below the 170 V limit.	DISCONNECTED / UNDERVOLTAGE, relay off.	32.9 V (readings 31.2, 30.0, 32.9 as the pot settled). LCD: TRIP UV / relayOFF. Terminal: FAULT:UNDERVOLTAGE. PASS.
PDF p.5 04_overcurrent_trip.png	04 Overcurrent	Current pot raised above the 10 A limit (17.56 A).	DISCONNECTED / OVERCURRENT, relay off.	Relay tripped (LCD: TRIP OV / relayOFF), but the fault reported is OVERVOLTAGE, not OVERCURRENT: voltage was 251.9 V, just over the 250 V limit, so OV took priority. Load is disconnected as required; this capture does not isolate the OC path. Re-take with voltage inside the window to show FAULT:OVERCURRENT.
PDF p.6 05_cooldown_recovery.png	05 Cooldown after fault clears	Fault (UV, 74.5 V) removed by moving the voltage pot back into range.	COOLDOWN with relay off before returning to MONITORING.	Terminal: UNDERVOLTAGE at 74.5 V, then STATE:COOLDOWN FAULT:NONE at 248 V and 280 V. LCD: COOLDOWN... Relay LED off during cooldown. PASS for entering cooldown with relay held off. The return to MONITORING is not visible in this capture.
PDF p.7 06a_reset_button_cooldown.png	06 Manual reset button (PB12), press	OV fault active (280 V, 17.56 A); RESET button pressed.	COOLDOWN after the press.	RESET highlighted. Terminal: DISCONNECTED / OVERVOLTAGE lines, then STATE:COOLDOWN FAULT:NONE. LCD: COOLDOWN... Relay off. PASS.
PDF p.8 06b_reset_button_refault.png	06 Manual reset button, outcome	Same run, ~2.4 s later; 280 V still applied.	Fault re-detected after cooldown because the condition persists.	Terminal: COOLDOWN, then back to STATE:DISCONNECTED FAULT:OVERVOLTAGE. LCD: TRIP OV / relayOFF. Shows reset does not override a persistent fault. PASS.
PDF p.9 07_uart_ov_setting.png	07 Protection setting changed (UART / settings)	Run at 199.4 V, 4.38 A with the OV limit changed from the 250 V default.	Status line reports the new limit; no false trip at 199.4 V.	Terminal: OV:265V UV:170V OC:10.0A, STATE:MONITORING FAULT:NONE. LCD: OK / relay ON. Relay on. The new limit (265 V) is visible, but the capture does not show how it was set (UART command or on-device buttons). Two status lines have a garbled voltage (0.?V), likely serial glitches. README lists UART RX as not verified in Wokwi, so treat this as supporting evidence only.

Wokwi Simulator | 01_normal.scenario.yaml U

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Problems Output Debug Console Terminal Ports Spell Checker Serial Monitor

Terminal

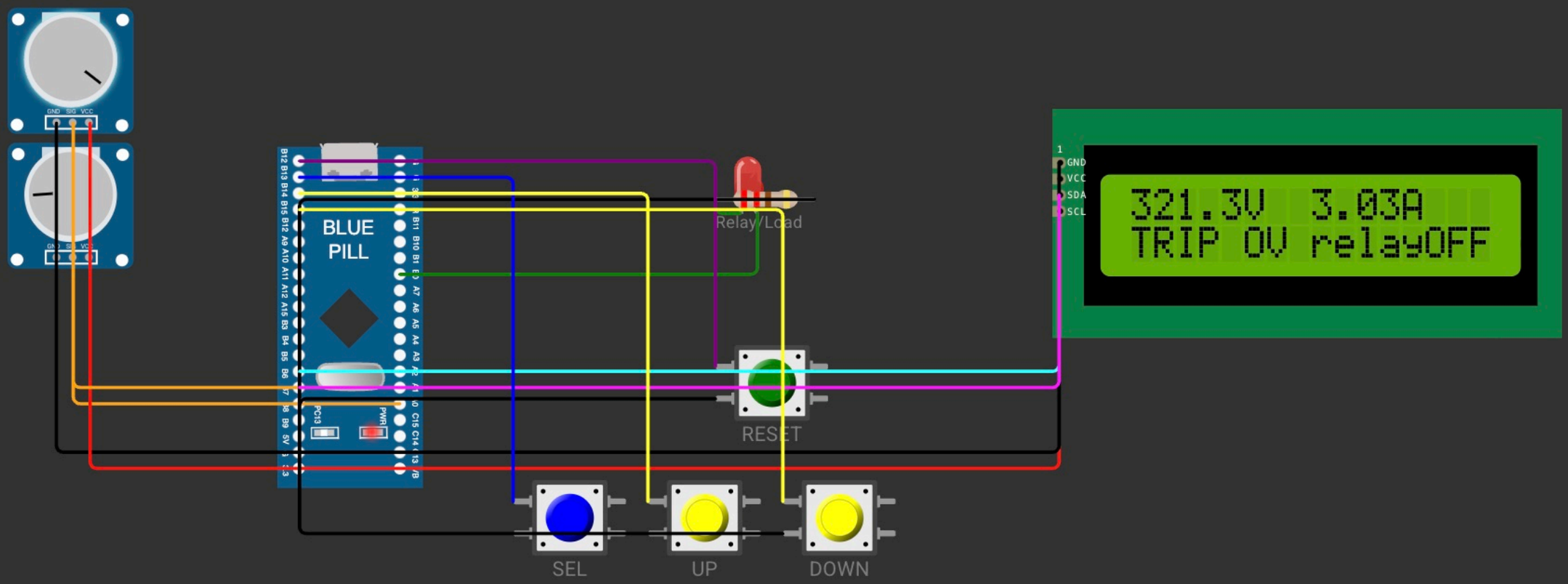
```
[STATUS] VOLT:188.1V CURR:3.03A STATE:MONITORING FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:188.1V CURR:3.03A STATE:MONITORING FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:188.1V CURR:3.03A STATE:MONITORING FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:188.1V CURR:3.03A STATE:MONITORING FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:188.1V CURR:3.03A STATE:MONITORING FAULT:NONE OV:250V UV:170V OC:10.0A
```

Wokwi Simulator | 02_overvoltage.scenario.yaml U

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Problems 2 | Output | Debug Console | Terminal | Ports | Spell Checker 2 | Serial Monitor

Terminal

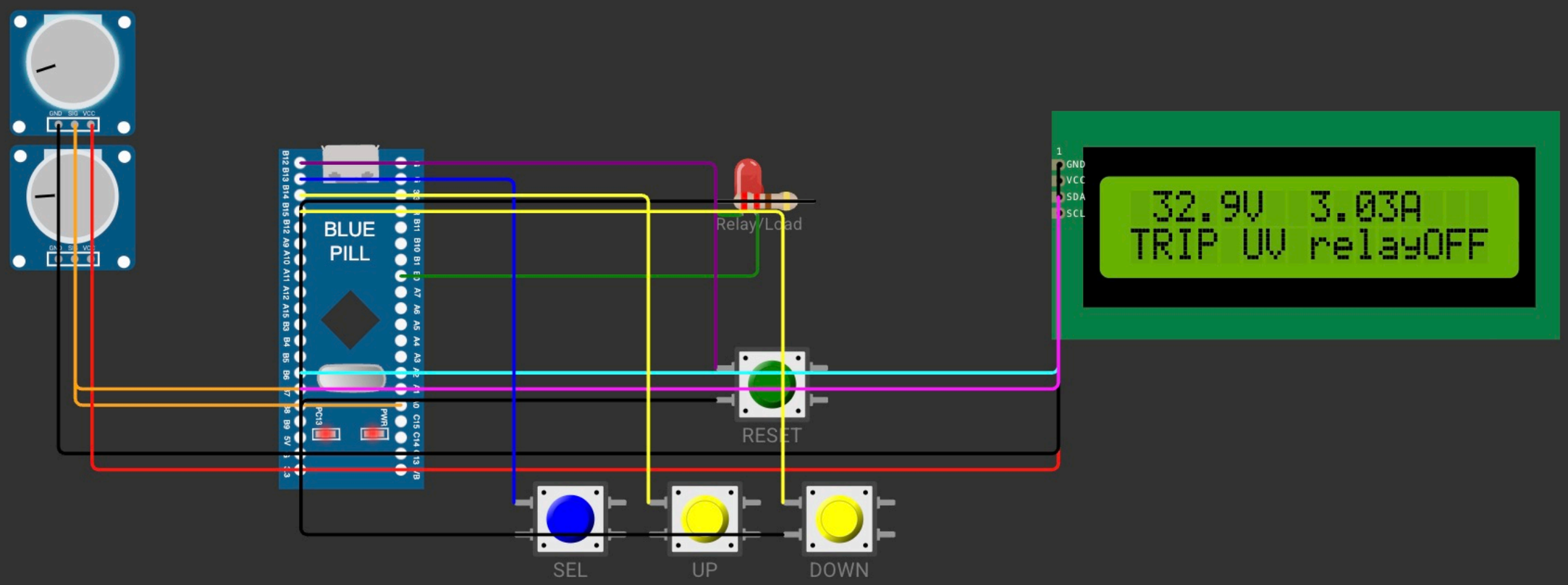
```
[STATUS] VOLT:321.3V CURR:3.03A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:321.3V CURR:3.03A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:-,)V UV:170V OC:10.0A
[STATUS] VOLT:321.3V CURR:3.03A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:321.3V CURR:3.03A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:321.3V CURR:3.03A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
```

Wokwi Simulator | 03_undervoltage.scenario.yaml U

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Problems 2 | Output | Debug Console | Terminal | Ports | Spell Checker 2 | Serial Monitor

Terminal

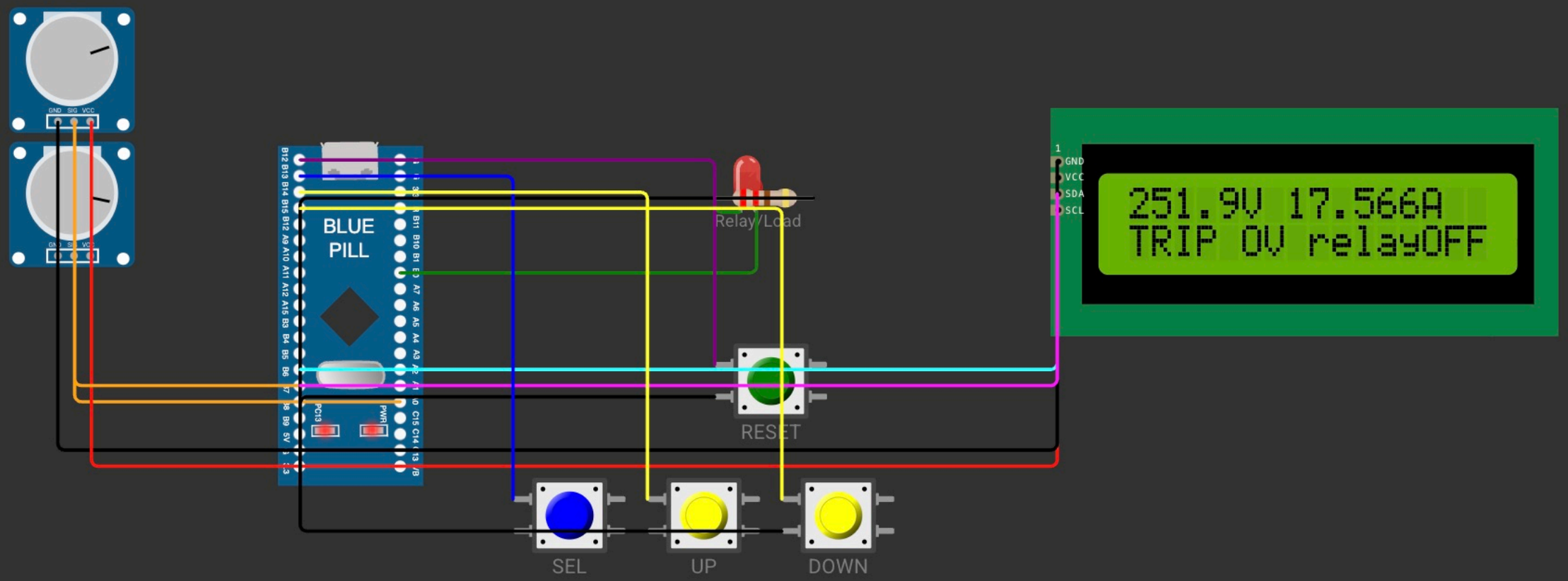
```
[STATUS] VOLT:31.2V CURR:3.03A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:30.0V CURR:3.03A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:32.9V CURR:3.03A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:32.9V CURR:3.03A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:32.9V CURR:3.03A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:32.9V CURR:3.03A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
```

Wokwi Simulator | 04_overcurrent.scenario.yaml U

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Problems 2 | Output | Debug Console | Terminal | Ports | Spell Checker 2 | Serial Monitor

Terminal

```
[STATUS] VOLT:251.9V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:251.9V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:251.9V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:251.9V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:251.9V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:251.9V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
```

deviceCare

Update

Explorer

- deviceCare
  - schematics
  - src
    - .settings
    - .vscode
    - build
    - cmake
    - Core
    - Drivers
      - CMSIS
      - STM32F1xx_HAL_Driv...
    - .clangd
    - .mxproject
    - .project
    - CMakeLists.txt
    - CMakePresets.json
    - startup_stm32f103x6.s
    - STM32F103XX_FLASH.ld
  - test_evidence
    - logs
    - scenarios
      - 01_normal.scenari... U
      - 02_overvoltage.sc... U
      - 03_undervoltage.s... U
      - 04_overcurrent.sc... U
      - 05_cooldown.scen... U
      - 06_reset_button.s... U
      - 07_uart.scenario.y... U
      - 08_setting_effect.... U
      - 09_timing.scenari... U

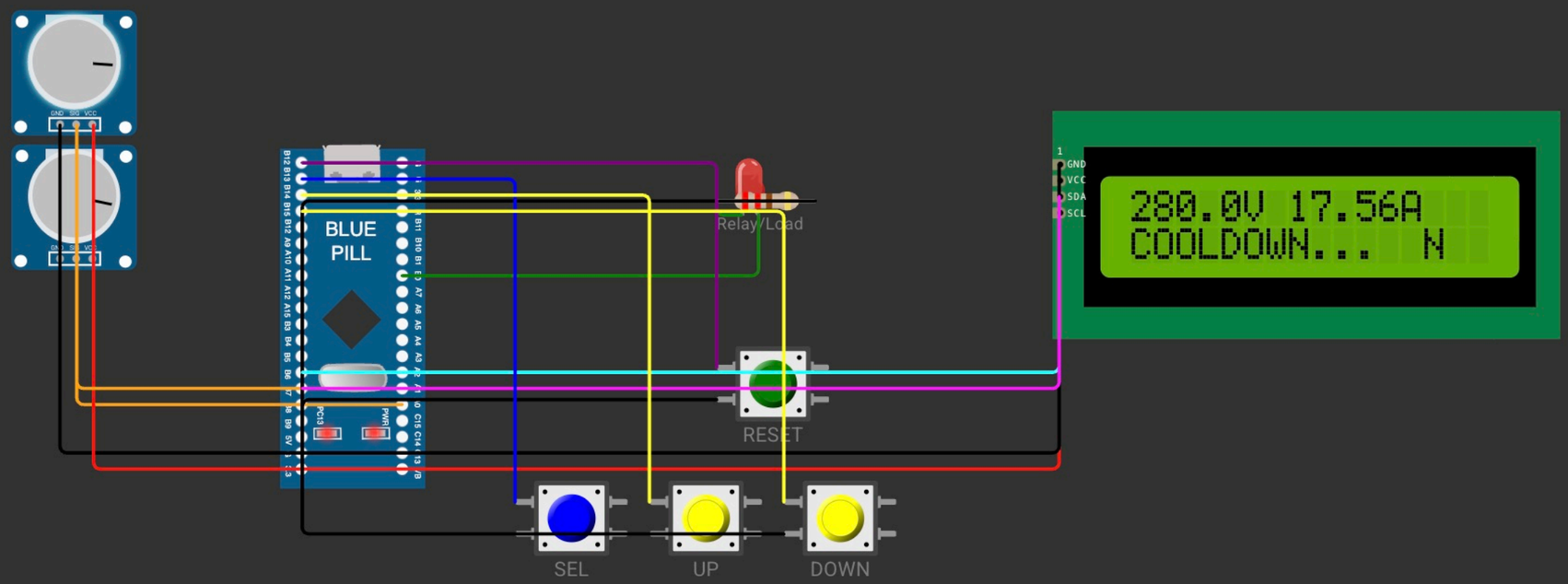
Outline

Timeline

Wokwi Simulator | 05_cooldown.scenario.yaml U

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Problems 1 | Output | Debug Console | Terminal | Ports | Spell Checker 1 | Serial Monitor

Terminal

```
[STATUS] VOLT:74.5V CURR:17.56A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:74.5V CURR:17.56A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:74.5V CURR:17.56A STATE:DISCONNECTED FAULT:UNDERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:248.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
```

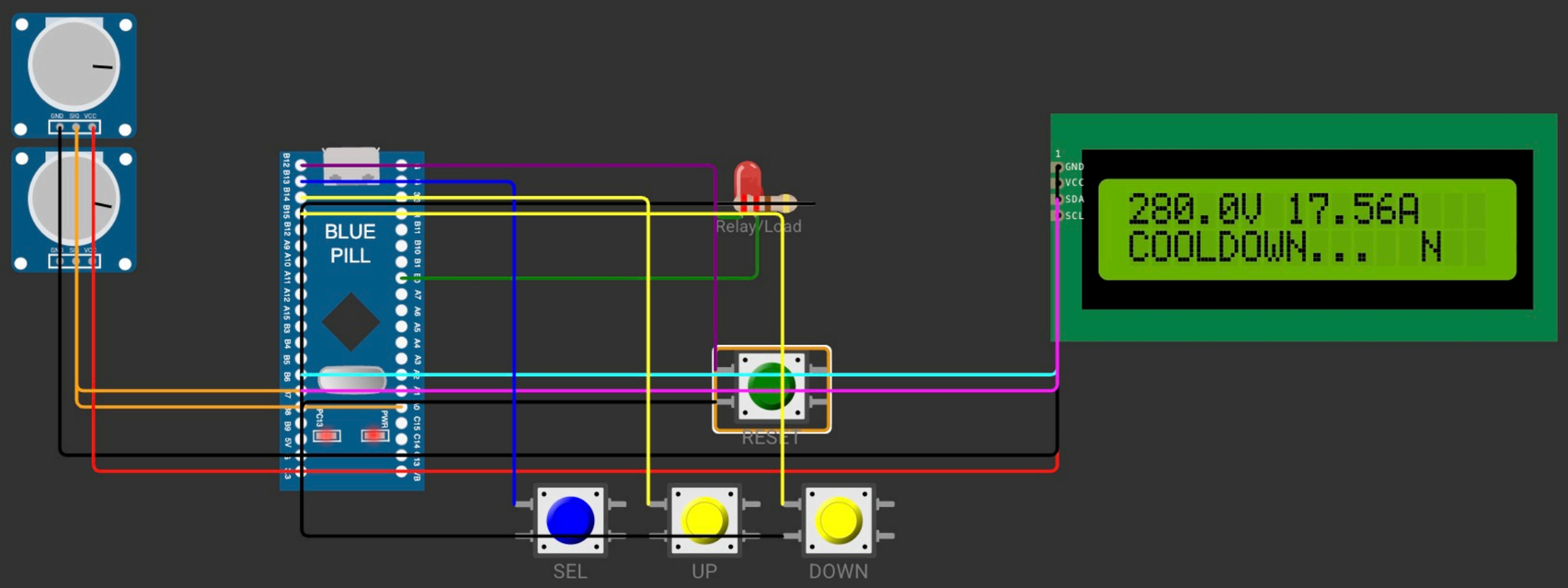
Explorer

- deviceCare
  - schematics
  - src
    - .settings
    - .vscode
    - build
    - cmake
    - Core
    - Drivers
      - CMSIS
      - STM32F1xx_HAL_Driv...
    - .clangd
    - .mxproject
    - .project
    - CMakeLists.txt
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    - startup_stm32f103x6.s
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      - 01_normal.scenari... U
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      - 03_undervoltage.s... U
      - 04_overcurrent.sc... U
      - 05_cooldown.scen... U
      - 06_reset_button.s... U
      - 07_uart.scenario.y... U
      - 08_setting_effect.... U
      - 09_timing.scenari... U
- Outline
- Timeline

Wokwi Simulator | 06_reset_button.scenario.yaml U

WOKWI Simulator

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Problems 1 | Output | Debug Console | Terminal | Ports | Spell Checker 1 | Serial Monitor

Terminal

```
[STATUS] VOLT:280.0V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.52A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
```

deviceCare

Update

Explorer

- deviceCare
  - schematics
  - src
    - .settings
    - .vscode
    - build
    - cmake
    - Core
    - Drivers
      - CMSIS
      - STM32F1xx_HAL_Driv...
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  - test_evidence
    - logs
    - scenarios
      - 01_normal.scenari... U
      - 02_overvoltage.sc... U
      - 03_undervoltage.s... U
      - 04_overcurrent.sc... U
      - 05_cooldown.scen... U
      - 06_reset_button.s... U
      - 07_uart.scenario.y... U
      - 08_setting_effect.... U
      - 09_timing.scenari... U

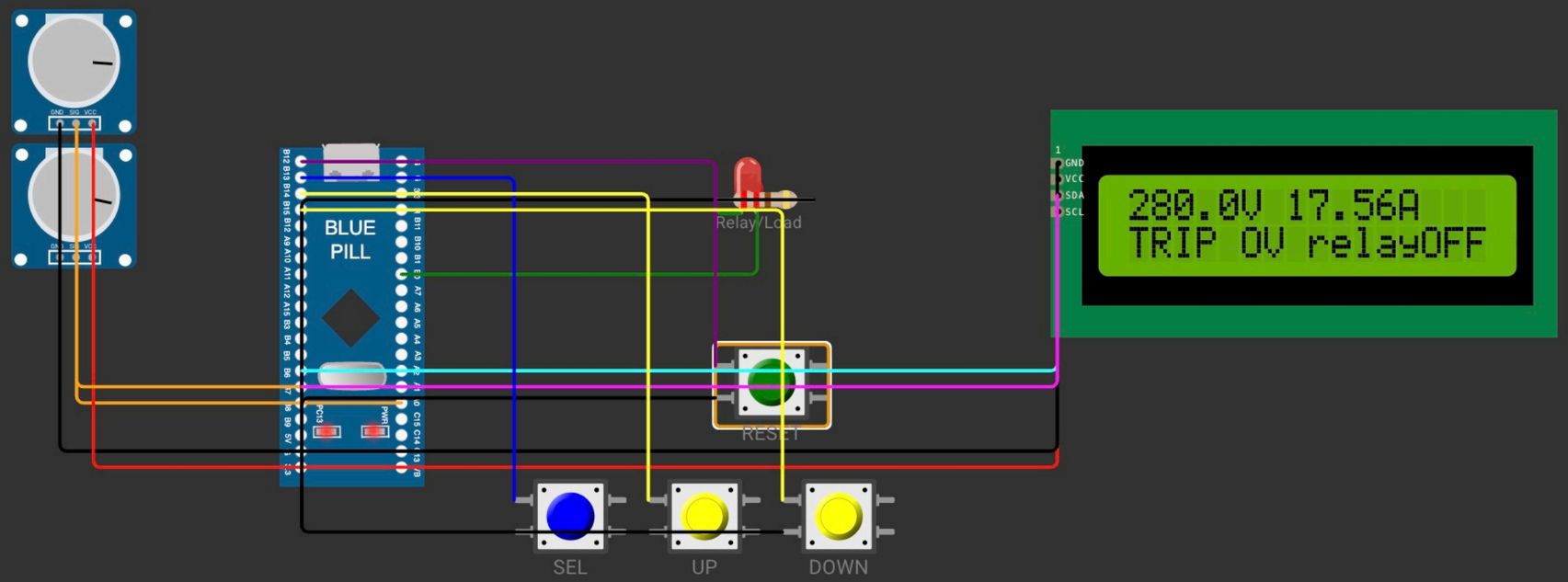
Outline

Timeline

Wokwi Simulator | 06_reset_button.scenario.yaml U

WOKWI Simulator

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Problems 1 | Output | Debug Console | Terminal | Ports | Spell Checker 1 | Serial Monitor

Terminal

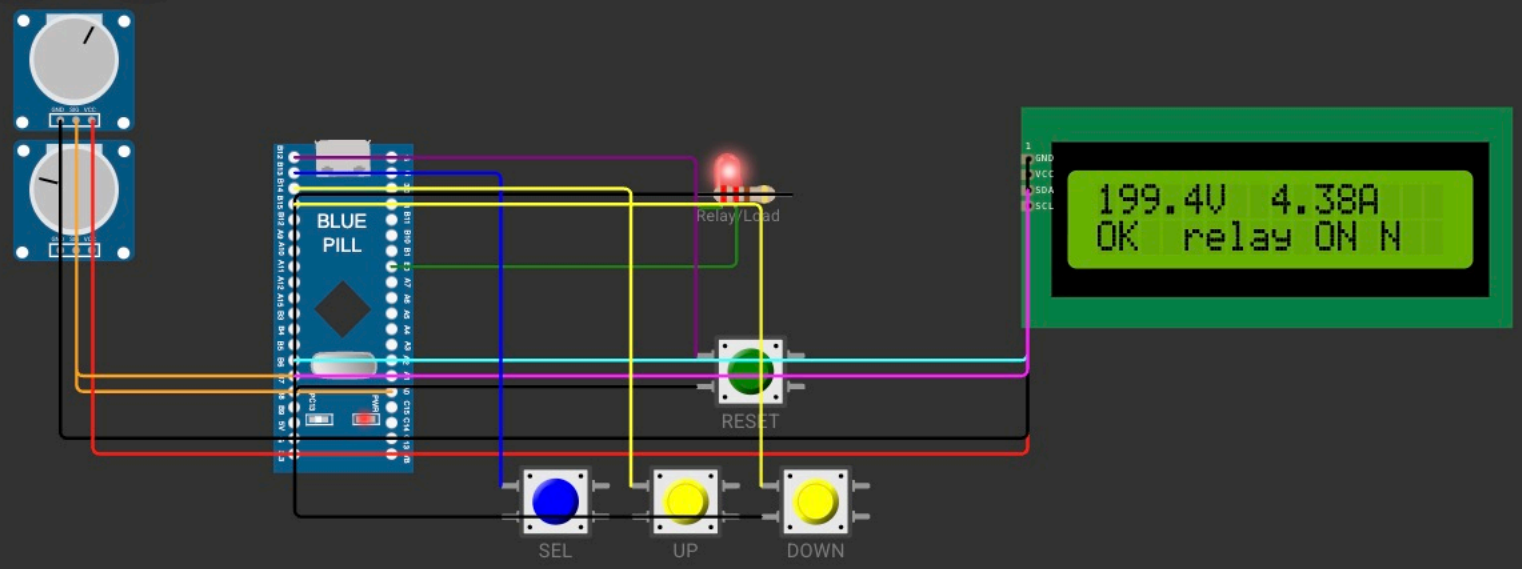
```
[STATUS] VOLT:280.0V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:COOLDOWN FAULT:NONE OV:250V UV:170V OC:10.0A
[STATUS] VOLT:280.0V CURR:17.56A STATE:DISCONNECTED FAULT:OVERVOLTAGE OV:250V UV:170V
```

07_uart.scenario.yaml U | Wokwi Simulator

WOKWI Simulator

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Problems 1 | Output | Debug Console | Terminal | Ports | Spell Checker 1 | Serial Monitor

Terminal

```
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:0.0V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.00A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:199.4V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
[STATUS] VOLT:0.0V CURR:4.38A STATE:MONITORING FAULT:NONE OV:265V UV:170V OC:10.0A
```